

```
pip install pandas
```

Requirement already satisfied: pandas in c:\users\tisht\anaconda3\lib\site-packages (2.3.3)

Requirement already satisfied: numpy>=1.26.0 in c:\users\tisht\anaconda3\lib\site-packages (from pandas) (2.3.5)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\tisht\anaconda3\lib\site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\tisht\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in c:\users\tisht\anaconda3\lib\site-packages (from pandas) (2025.2)

Requirement already satisfied: six>=1.5 in c:\users\tisht\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

Note: you may need to restart the kernel to use updated packages.

```
pip install matplotlib
```

Requirement already satisfied: matplotlib in c:\users\tisht\anaconda3\lib\site-packages (3.10.6)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (1.3.3)

Requirement already satisfied: cycler>=0.10 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (4.60.1)

Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (1.4.9)

Requirement already satisfied: numpy>=1.23 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (2.3.5)

Requirement already satisfied: packaging>=20.0 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (25.0)

Requirement already satisfied: pillow>=8 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (12.0.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (3.2.5)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)

Requirement already satisfied: six>=1.5 in c:\users\tisht\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

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```
pip install seaborn
```

Requirement already satisfied: seaborn in c:\users\tisht\anaconda3\lib\site-packages (0.13.2)

Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\tisht\anaconda3\lib\site-packages (from seaborn) (2.3.5)

Requirement already satisfied: pandas>=1.2 in c:\users\tisht\anaconda3\lib\site-packages (from seaborn) (2.3.3)

Requirement already satisfied: matplotlib!=3.6.1,>=3.4 in c:\users\tisht\anaconda3\lib\site-packages (from seaborn) (3.10.6)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\tisht\anaconda3\lib\site-packages (from matplotlib!=3.6.1,>=3.4->seaborn) (1.3.3)

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Note: you may need to restart the kernel to use updated packages.

```
pip install numpy
```

Requirement already satisfied: numpy in c:\users\tisht\anaconda3\lib\site-packages (2.3.5)

Note: you may need to restart the kernel to use updated packages.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv('Customer Churn.csv')
df
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	\
0	7590-VHVEG	Female	0	Yes	No	1	

1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2
...	...	...	...	...	...	...
7038	6840-RESVB	Male	0	Yes	Yes	24
7039	2234-XADUH	Female	0	Yes	Yes	72
7040	4801-JZAZL	Female	0	Yes	Yes	11
7041	8361-LTMKD	Male	1	Yes	No	4
7042	3186-AJIEK	Male	0	No	No	66

	PhoneService	MultipleLines	InternetService
OnlineSecurity	...	\	
0	No	No phone service	DSL
No	...		
1	Yes	No	DSL
Yes	...		
2	Yes	No	DSL
Yes	...		
3	No	No phone service	DSL
Yes	...		
4	Yes	No	Fiber optic
No	...		
...	...	...	...
.			
7038	Yes	Yes	DSL
Yes	...		
7039	Yes	Yes	Fiber optic
No	...		
7040	No	No phone service	DSL
Yes	...		
7041	Yes	Yes	Fiber optic
No	...		
7042	Yes	No	Fiber optic
Yes	...		

	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
Contract	\			
0	No	No	No	No
to-month				
1	Yes	No	No	No
One year				
2	No	No	No	No
to-month				
3	Yes	Yes	No	No
One year				
4	No	No	No	No
to-month				
...	...	...	...	...

...				
7038	Yes	Yes	Yes	Yes
One year				
7039	Yes	No	Yes	Yes
One year				
7040	No	No	No	No Month-
to-month				
7041	No	No	No	No Month-
to-month				
7042	Yes	Yes	Yes	Yes
Two year				

	PaperlessBilling	PaymentMethod	MonthlyCharges
TotalCharges \			
0	Yes	Electronic check	29.85
29.85			
1	No	Mailed check	56.95
1889.5			
2	Yes	Mailed check	53.85
108.15			
3	No	Bank transfer (automatic)	42.30
1840.75			
4	Yes	Electronic check	70.70
151.65			

...	...	...	...
...			
7038	Yes	Mailed check	84.80
1990.5			
7039	Yes	Credit card (automatic)	103.20
7362.9			
7040	Yes	Electronic check	29.60
346.45			
7041	Yes	Mailed check	74.40
306.6			
7042	Yes	Bank transfer (automatic)	105.65
6844.5			

	Churn
0	No
1	No
2	Yes
3	No
4	Yes
...	...
7038	No
7039	No
7040	No
7041	Yes
7042	No

[7043 rows x 21 columns]

df.head()

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure
0	7590-VHVEG	Female	0	Yes	No	1
1	5575-GNVDE	Male	0	No	No	34
2	3668-QPYBK	Male	0	No	No	2
3	7795-CF0CW	Male	0	No	No	45
4	9237-HQITU	Female	0	No	No	2

	MultipleLines	InternetService	OnlineSecurity	...
0	No phone service	DSL	No	...
1	No	DSL	Yes	...
2	No	DSL	Yes	...
3	No phone service	DSL	Yes	...
4	No	Fiber optic	No	...

	TechSupport	StreamingTV	StreamingMovies	Contract
0	No	No	No	Month-to-month
1	No	No	No	One year
2	No	No	No	Month-to-month
3	Yes	No	No	One year
4	No	No	No	Month-to-month

	PaymentMethod	MonthlyCharges	TotalCharges	Churn
0	Electronic check	29.85	29.85	No
1	Mailed check	56.95	1889.5	No
2	Mailed check	53.85	108.15	Yes
3	Bank transfer (automatic)	42.30	1840.75	No
4	Electronic check	70.70	151.65	Yes

```
[5 rows x 21 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 7043 entries, 0 to 7042
```

```
Data columns (total 21 columns):
```

#	Column	Non-Null	Count	Dtype
0	customerID	7043	non-null	object
1	gender	7043	non-null	object
2	SeniorCitizen	7043	non-null	int64
3	Partner	7043	non-null	object
4	Dependents	7043	non-null	object
5	tenure	7043	non-null	int64
6	PhoneService	7043	non-null	object
7	MultipleLines	7043	non-null	object
8	InternetService	7043	non-null	object
9	OnlineSecurity	7043	non-null	object
10	OnlineBackup	7043	non-null	object
11	DeviceProtection	7043	non-null	object
12	TechSupport	7043	non-null	object
13	StreamingTV	7043	non-null	object
14	StreamingMovies	7043	non-null	object
15	Contract	7043	non-null	object
16	PaperlessBilling	7043	non-null	object
17	PaymentMethod	7043	non-null	object
18	MonthlyCharges	7043	non-null	float64
19	TotalCharges	7043	non-null	object
20	Churn	7043	non-null	object

```
dtypes: float64(1), int64(2), object(18)
```

```
memory usage: 1.1+ MB
```

```
#Replacing blanks with 0 as tenure is 0 and no total charges are recorded
```

```
df["TotalCharges"]=df["TotalCharges"].replace(" ", "0")
```

```
df["TotalCharges"]=df["TotalCharges"].astype("float")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 7043 entries, 0 to 7042
```

```
Data columns (total 21 columns):
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#	Column	Non-Null	Count	Dtype
0	customerID	7043	non-null	object
1	gender	7043	non-null	object
2	SeniorCitizen	7043	non-null	int64
3	Partner	7043	non-null	object

4	Dependents	7043	non-null	object
5	tenure	7043	non-null	int64
6	PhoneService	7043	non-null	object
7	MultipleLines	7043	non-null	object
8	InternetService	7043	non-null	object
9	OnlineSecurity	7043	non-null	object
10	OnlineBackup	7043	non-null	object
11	DeviceProtection	7043	non-null	object
12	TechSupport	7043	non-null	object
13	StreamingTV	7043	non-null	object
14	StreamingMovies	7043	non-null	object
15	Contract	7043	non-null	object
16	PaperlessBilling	7043	non-null	object
17	PaymentMethod	7043	non-null	object
18	MonthlyCharges	7043	non-null	float64
19	TotalCharges	7043	non-null	float64
20	Churn	7043	non-null	object

dtypes: float64(2), int64(2), object(17)

memory usage: 1.1+ MB

df.isnull().sum()

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

df.describe()

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304

std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

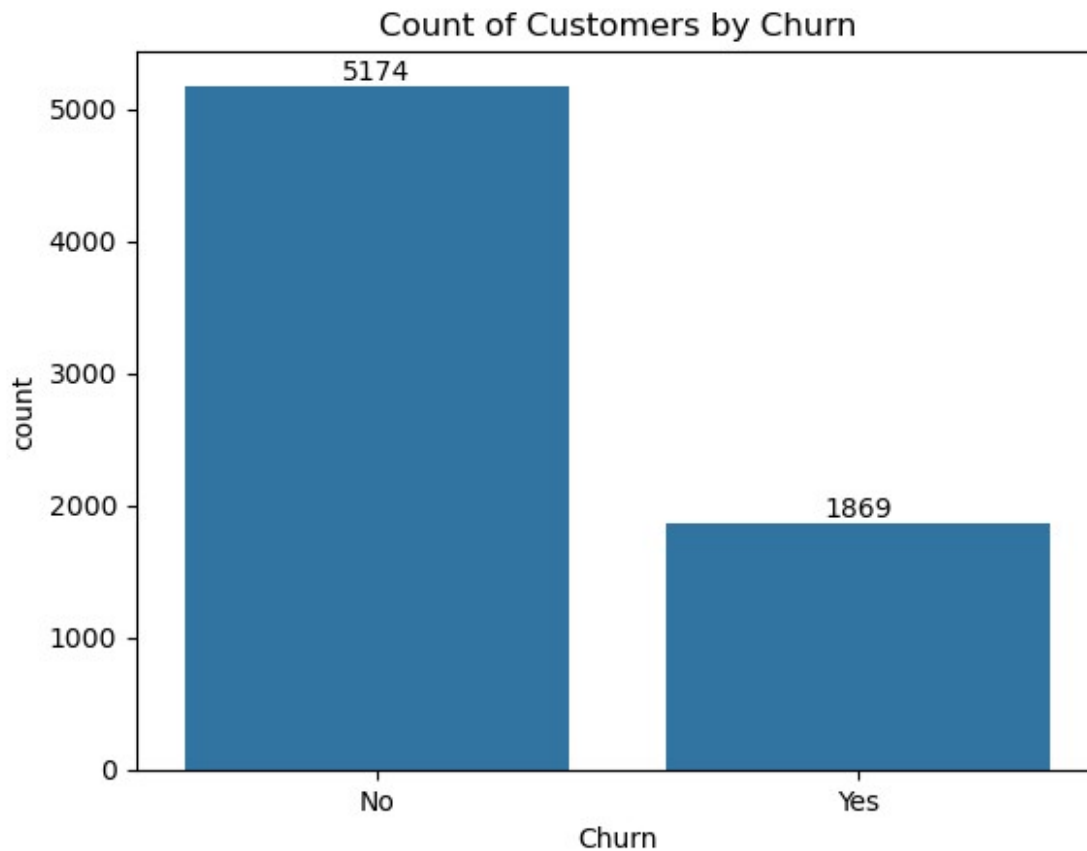
```
def conv(value):
    if value == 1:
        return "yes"
    else:
        return "no"
```

```
df['SeniorCitizen'] = df['SeniorCitizen'].apply(conv)
```

#converted 0 and 1 values of senior citizen to yes/ no to make it easir to understand

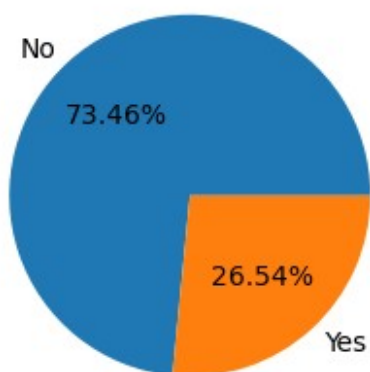
```
ax = sns.countplot(x='Churn', data = df)
ax.bar_label(ax.containers[0])
plt.title("Count of Customers by Churn")
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```

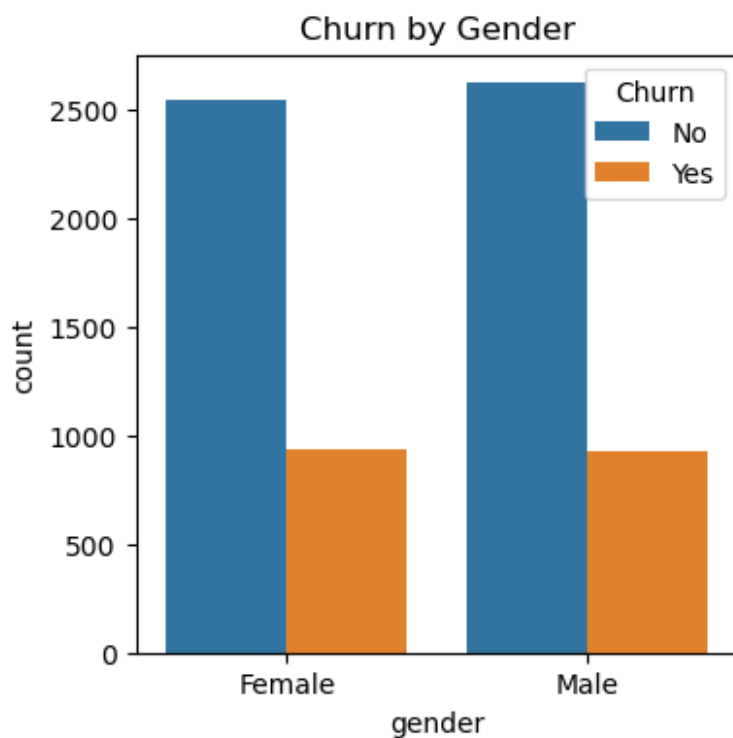
```
plt.figure(figsize=(3,4))
gb = df.groupby("Churn").agg({'Churn':"count"})
plt.pie(gb['Churn'], labels = gb.index, autopct = "%1.2f%%")
plt.title("Percentage of Churned Customers", fontsize = 10)
plt.show()
```

Percentage of Churned Customers

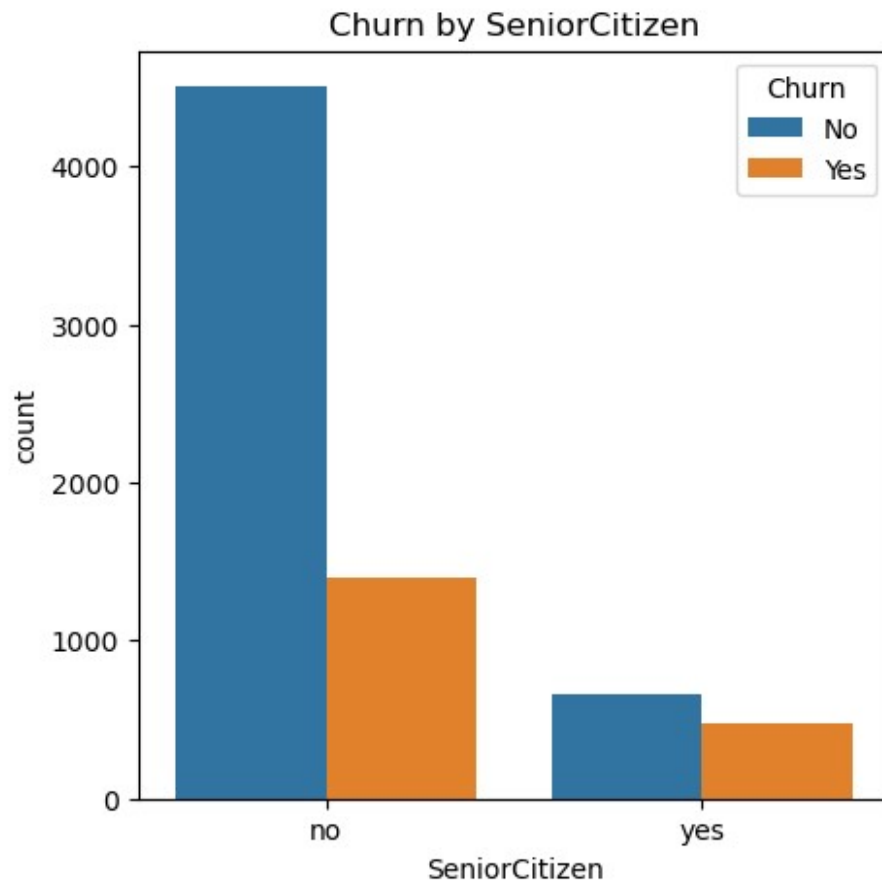


From the give Pie Chart we can conclude that 26.54% of our customers have churned out. Now let's explore the reason behind it

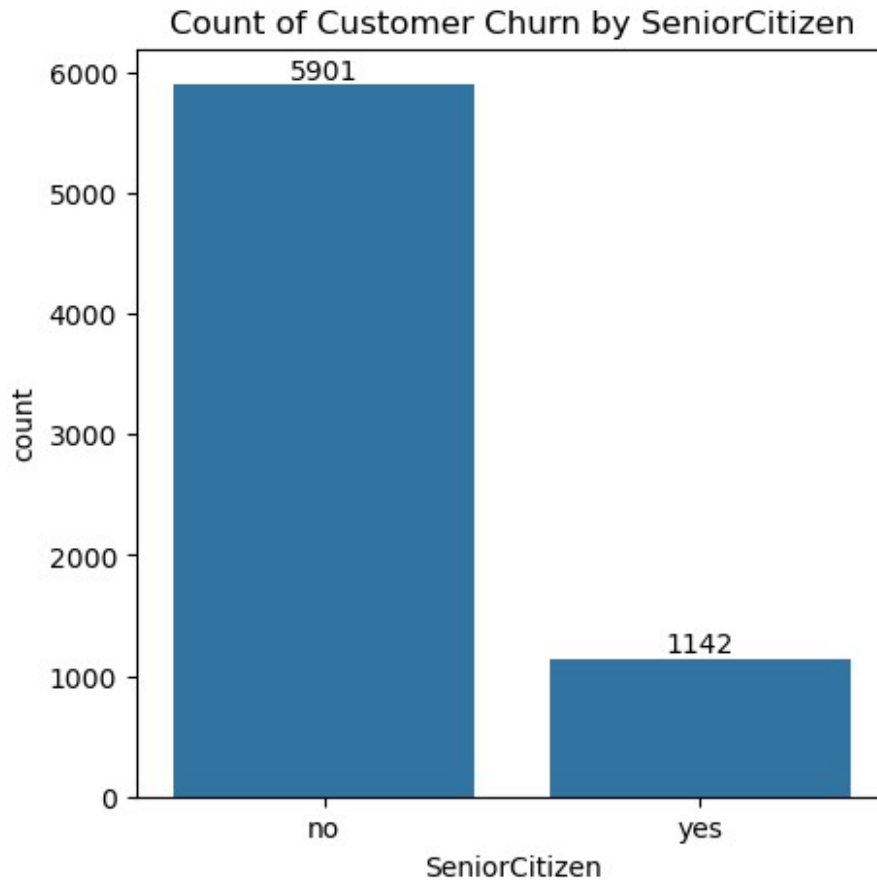
```
plt.figure(figsize = (4,4))
sns.countplot(x = "gender", data = df, hue = "Churn")
plt.title('Churn by Gender')
plt.show()
```



```
plt.figure(figsize = (5,5))
sns.countplot(x = "SeniorCitizen", data = df, hue = "Churn")
plt.title('Churn by SeniorCitizen')
plt.show()
```



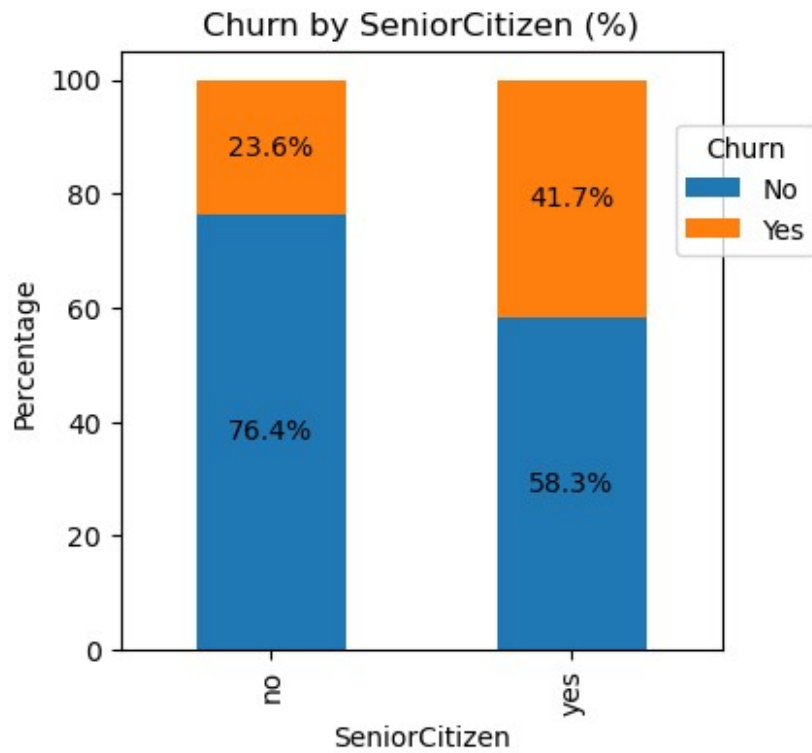
```
plt.figure(figsize = (5,5))  
ax = sns.countplot(x = "SeniorCitizen", data = df)  
ax.bar_label(ax.containers[0])  
plt.title('Count of Customer Churn by SeniorCitizen')  
plt.show()
```



```
ct = pd.crosstab(df['SeniorCitizen'], df['Churn'], normalize='index')
* 100
ax = ct.plot(kind='bar', stacked=True, figsize=(4,4))

for i, row in enumerate(ct.values):
    cumulative = 0
    for j, val in enumerate(row):
        plt.text(i, cumulative + val/2, f'{val:.1f}%', ha='center',
va='center')
        cumulative += val

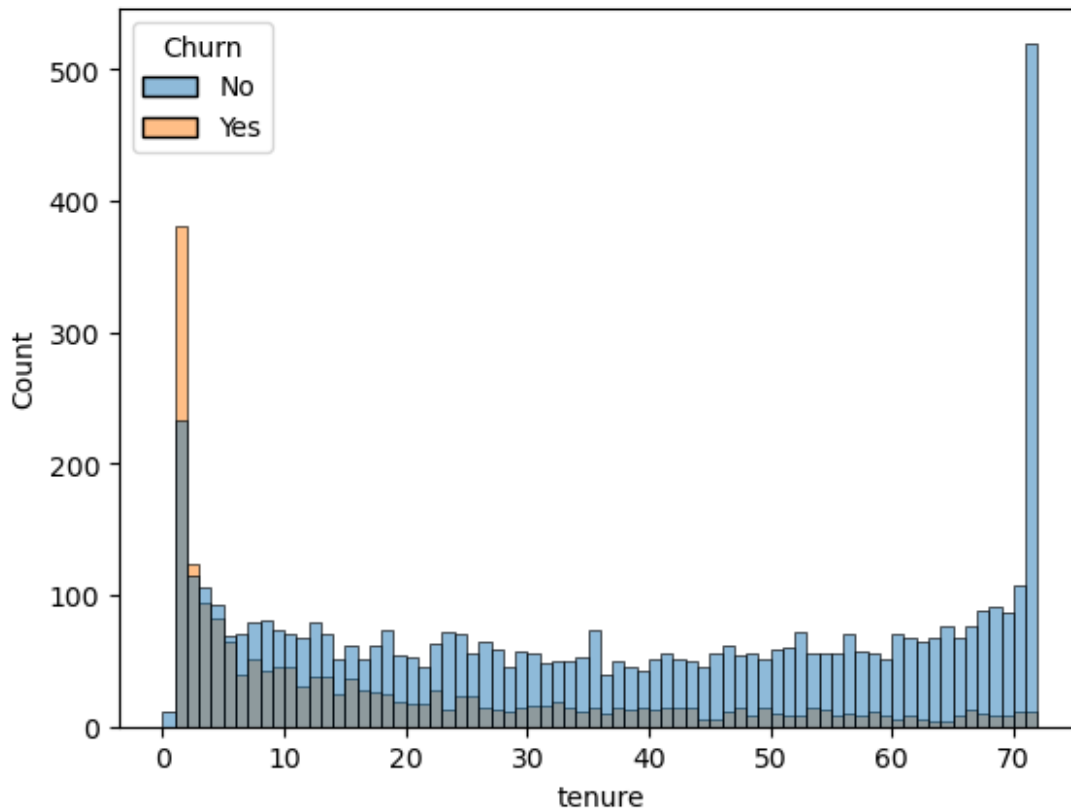
plt.title('Churn by SeniorCitizen (%)')
plt.ylabel('Percentage')
plt.xlabel('SeniorCitizen')
plt.legend(title='Churn', bbox_to_anchor = (0.9,0.9))
plt.figure(figsize = (5,5))
plt.show()
```



<Figure size 500x500 with 0 Axes>

comparative a greater percentage of people in senior citizen category have churned

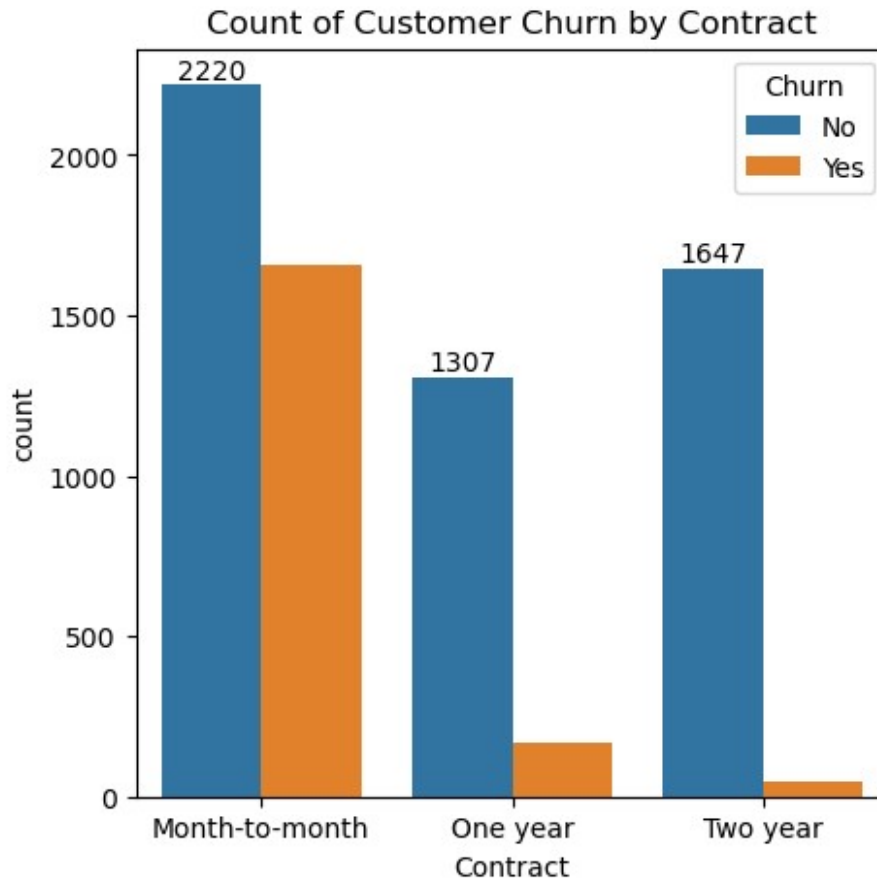
```
sns.histplot(x = 'tenure', data = df, bins = 72, hue = 'Churn')  
plt.figure(figsize = (9,4))  
plt.show()
```



<Figure size 900x400 with 0 Axes>

People who have used our services for long time have stayed and people who have used our services for 1 or 2 months have churned

```
plt.figure(figsize = (5,5))
ax = sns.countplot(x = "Contract", data = df, hue = 'Churn')
ax.bar_label(ax.containers[0])
plt.title('Count of Customer Churn by Contract')
plt.show()
```



#people who have month to month contract are likely to churn then those who have one or two years contract

```
df.columns.values
```

```
array(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents', 'tenure', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'], dtype=object)
```

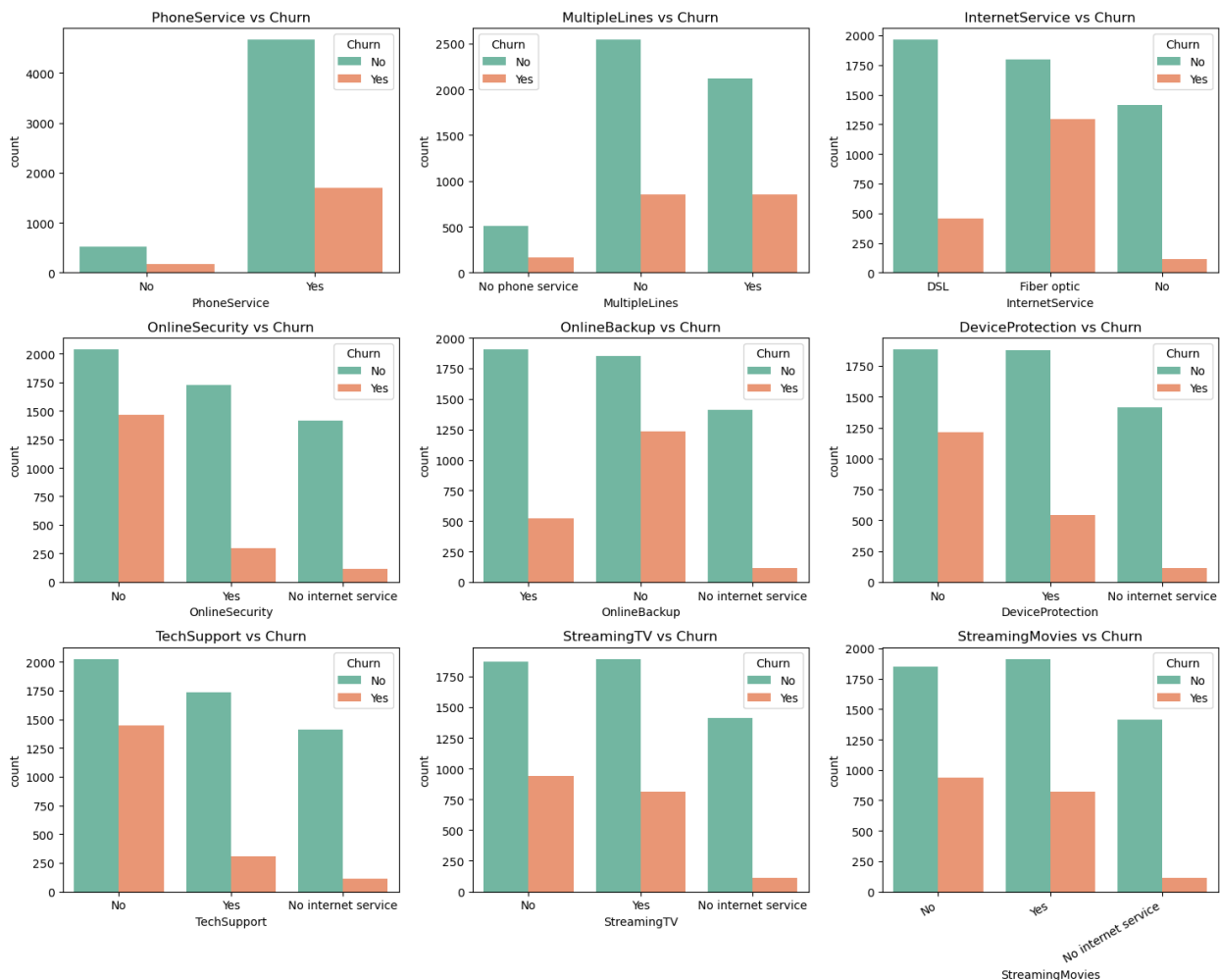
```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cols = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]
```

Set figure size

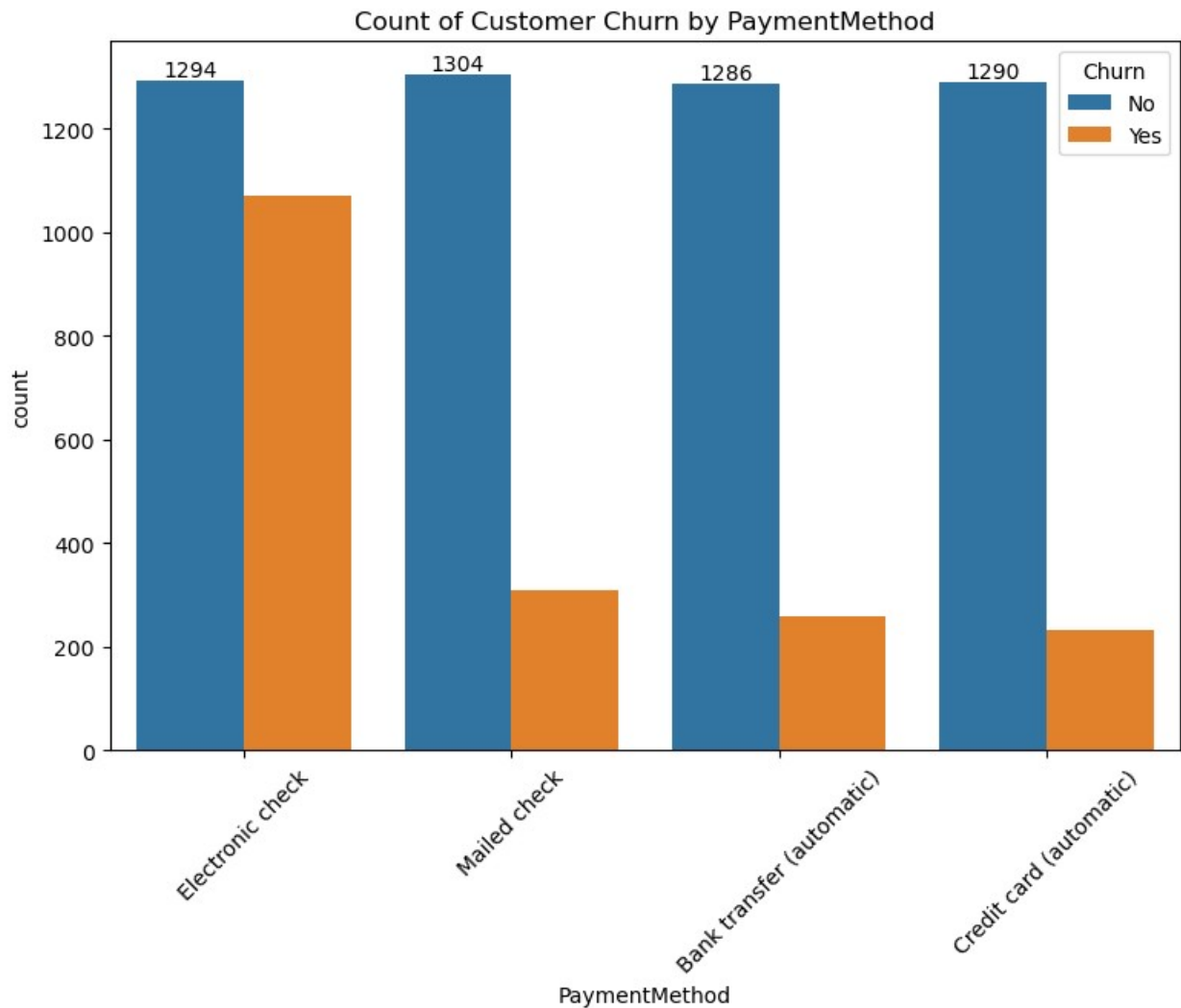
```
plt.figure(figsize=(15, 12))

for i, col in enumerate(cols, 1):
    plt.subplot(3, 3, i)
    sns.countplot(data=df, x=col, hue='Churn', palette='Set2')
    plt.title(f'{col} vs Churn')
    plt.xticks(rotation=0)
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.show()
```



Customers with **no add-on services** (like Online Security, Tech Support, Device Protection) show significantly higher churn compared to those who have these services. Users subscribed to **fiber optic internet** tend to churn more than DSL users. Customers without **backup, security, or support features** are more likely to leave, indicating these services improve retention. Overall, **value-added services play a key role in reducing churn**, while basic service users are at higher risk.


```
plt.figure(figsize = (9,6))
ax = sns.countplot(x = "PaymentMethod", data = df, hue="Churn")
ax.bar_label(ax.containers[0])
plt.title("Count of Customer Churn by PaymentMethod")
plt.xticks(rotation=45)
plt.show()
```



Customer is likely to churn when he is using electronic check as a payment.